

Team WeightWatchers: Testing the Tamper-Resistance of Open-Weight LLMs

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Abstract

Recent advancements in open-weight Large Language Models (LLMs) have necessitated the development of tampering-resistant safeguards to prevent malicious actors from removing safety alignment. The state-of-the-art method, Tampering Attack Resistance (TAR), utilizes meta-learning to robustly unlearn hazardous knowledge. In this work, we investigate the mechanistic underpinnings of TAR and demonstrate that its protection is mediated by a single, linear “refusal direction” in the residual stream. We apply the refusal direction ablation technique proposed by Arditì et al. (2024) [1] to successfully bypass TAR safeguards with greater than 85% attack success rate, outperforming expensive fine-tuning baselines while requiring orders of magnitude less compute. Conversely, we find that neuron-level interventions are insufficient to invert safety behaviors, suggesting that safety in TAR models is encoded as a distributed linear feature rather than through sparse neuronal circuits. Our findings highlight a critical vulnerability in current robust alignment techniques: they harden models against gradient-based optimization but leave the underlying refusal representations intact and accessible.

1. Introduction

As open-weight Large Language Models (LLMs) become increasingly capable, the risk of dual-use—where models are repurposed for malicious tasks such as bioweapon design or cyberattacks—has grown significantly. Standard safety alignment techniques, such as Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF), are known to be brittle; adversaries can remove these safeguards with as few as 10 steps of fine-tuning.

In response, “durable” safeguards like Tampering Attack Resistance (TAR) have been proposed. TAR employs a bi-level optimization strategy to train models that resist safety removal even after hundreds of fine-tuning steps. However, the mechanistic basis of this resistance remains underexplored. Does TAR truly erase hazardous knowledge, or does it merely mask it?

In this paper, we answer this question by applying the “refusal direction” methodology introduced by Arditì et al. (2024) [1] to the TAR-hardened model. By identifying a linear vector in the model’s activation space responsible for refusal behavior and surgically ablating it, we demonstrate that TAR does not remove the underlying mechanism of refusal but rather fortifies the weights against standard gradient descent updates. We show that this existing technique can “jailbreak” the supposedly tamper-resistant model with-

out the need for expensive fine-tuning.

2. Tampering Attack Resistance (TAR)

The model studied in this work, **Llama3-8B-Instruct-TAR**, is built upon the concept of **Tampering Attack Resistance (TAR)** training. TAR is a meta-learning method designed to build robust safeguards into open-weight Large Language Models (LLMs) specifically against *tampering attacks*, which are defined as adversarial fine-tuning attempts that modify model weights to remove safety mechanisms.

Existing refusal and unlearning safeguards in open-weight models are typically brittle and can be trivially bypassed after only a few steps of fine-tuning on harmful data. TAR addresses this vulnerability by training the model to minimize a proxy safety loss *after* an adversarial fine-tuning step is applied. This approach forces the model to maintain its safety configuration even when an adversary attempts to revert it through optimization. The goal of TAR is to achieve robustness that can withstand hundreds of steps of adversarial fine-tuning while preserving the model’s benign general capabilities. The exact algorithm of TAR is present in Appendix B.

3. Methodology

3.1 Dataset Collection and Processing

Our analysis employed a comprehensive suite of eight primary datasets, segregated into sets of harmful and harmless instructions to facilitate the contrastive analysis necessary for direction identification. The datasets were standardized into a consistent JSON format and partitioned into disjoint training, validation, and testing splits. The primary harmful instruction datasets included: AdvBench, JailbreakBench, HarmBench, TDC2023, MaliciousInstruct, and StrongREJECT, and the harmless instruction dataset used was Alpaca. Capability and hazardous knowledge testing utilized Alpaca (benign instructions). This rigorous partitioning ensured that validation sets used for direction selection were separate from the final evaluation sets.

3.2 Refusal Direction Generation

The core of our methodology adapts the technique proposed by Arditì et al. (2024) to identify a *refusal direction*—a vector in the model’s residual stream activation space hypothesized to encode the model’s refusal behavior. We employed the widely used difference-in-means technique (Marks and Tegmark, 2023) to isolate this direction.

The process is formalized as follows:

1. We perform a forward pass of a training set of harmful instructions, caching the residual stream activations A_h .
2. We perform a forward pass of a training set of harmless instructions, caching the residual stream activations A_s .
3. For each layer l and token position p , we compute the mean activation for harmful inputs $\mu_h^{(l,p)}$ and harmless inputs $\mu_s^{(l,p)}$:

$$\mu_h^{(l,p)} = \text{mean}(A_h^{(l,p)}) \quad \text{and} \quad \mu_s^{(l,p)} = \text{mean}(A_s^{(l,p)})$$

4. The candidate refusal direction $r^{(l,p)}$ is calculated as the difference vector:

$$r^{(l,p)} = \mu_h^{(l,p)} - \mu_s^{(l,p)}$$

This process yields a tensor of candidate refusal directions of shape (n_positions, n_layers, d_{model}), where d_{model} is the residual stream dimension. This simple linear subtraction in the activation space is key to isolating the feature responsible for refusal.

3.3 Experimental Conditions

Our experiments involved three distinct conditions applied to the target model, **Llama3-8B-Instruct-TAR-Biov2**, to evaluate the necessity and sufficiency of the identified refusal direction:

1. **Baseline (No Intervention)**: This condition utilizes the original, unmodified **Llama3-8B-Instruct-TAR-Biov2** model. This serves as the control group, representing the model’s native safety performance, which is expected to exhibit strong Refusal behavior on harmful prompts.
2. **Ablation (Direction Removed)**: This is our primary intervention, designed to test the *necessity* of the refusal direction for safety behavior. The identified refusal direction r^* is **directionally ablated** from the model’s residual stream. This involves projecting out the component of the activation vector along r^* across all layers and token positions during inference. Mathematically, for a given activation x , the intervention is:

$$x' = x - \hat{r}\hat{r}^T x$$

where $\hat{r}$ is the unit-norm refusal direction. This is hypothesized to surgically disable the refusal mechanism.

3. **Activation Addition (ActAdd)**: This intervention tests the *sufficiency* of the refusal direction to induce the safety behavior. The direction vector r^* is **added** to the residual stream activations at its optimal source layer and all token positions during inference. This is expected to push activations toward the "refusal" state, inducing the model to refuse even benign, harmless instructions. The intervention is:

$$x^{(l^*)} \leftarrow x^{(l^*)} + r^*$$

3.4 Direction Selection Process

To identify the single, most causal direction r^* from the candidates, each vector was evaluated against three crucial metrics on validation datasets:

- **Ablation Refusal Score ($\downarrow$)**: Measures the reduction in refusal frequency on harmful prompts when the direction is ablated. Lower scores indicate greater efficacy in safety bypass.
- **Steering Score ($\uparrow$)**: Measures the degree to which adding the direction (ActAdd) induces refusal on harmless prompts, confirming the direction’s role as a causal mediator of refusal behavior.
- **KL Divergence ($\downarrow$)**: Measures the distribution shift between baseline and ablated outputs on harmless prompts. This metric ensures minimal functional damage to general capabilities.

Candidates were subject to strict filtering: directions from the last 20% of layers (to avoid proximity to unembedding layers) and those with a KL divergence exceeding a threshold of 0.1 were excluded. The final selected direction r^* was the one with the lowest Ablation Refusal Score among all filtered candidates.

3.5 Evaluation Framework

Three primary experimental conditions were tested: **Baseline** (no intervention), **Ablation** (refusal direction removed), and **ActAdd** (refusal direction added).

Performance was assessed using two methodologies:

- **Substring Matching (ASR)**: A rule-based classifier checking for common refusal phrases (e.g., "I’m sorry", "I cannot", "As an AI").
- **LlamaGuard2 (Safety ASR)**: A robust AI-powered safety classifier (meta-llama/Meta-Llama-Guard-2-8B) to detect 13 categories of harmful content, providing a more rigorous measure of Attack Success Rate (ASR).

These were applied across: JailbreakBench (primary ASR), Harmless instructions (False Positive Rate), WMDP-Bio (Hazardous Knowledge Accuracy), and The Pile/Alpaca (Capability Preservation via Cross-Entropy Loss and Perplexity).

4. Experiments and Results

4.1 Selected Refusal Direction

The rigorous selection process identified the optimal refusal direction r^* for the model at Position -5 (the 5th token from the end of the input sequence) and Layer 12 (a middle layer). The metrics for the selected direction, compared to the baseline, are summarized in Table 1. The significant reduction in the Refusal Score confirms the direction’s high causal efficacy.

Table 1: Metrics for the Selected Refusal Direction

Metric	Value	Note
Position	-5	5th token from end
Layer	12	Middle layer
Refusal Score (Ablation)	-7.51	(Baseline: 3.40)
Steering Score (ActAdd)	0.45	(Baseline: -9.77)
KL Divergence	0.016	Low value ensures minimal side effects

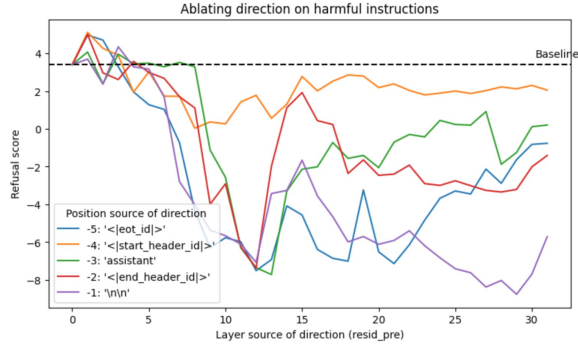


Figure 1: Finding the optimal refusal direction

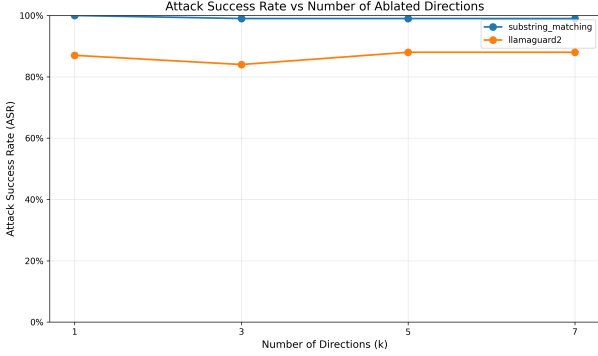


Figure 2: ASR vs No. of Ablated Direction

4.2 Choosing the Top- k Refusal Directions

We identify refusal-related features by ranking candidate directions using their refusal score, steering score, and KL divergence under activation ablation. The highest-ranked directions consistently appear in layers 11–13, with the strongest direction located at layer 12, position -5 . Figure 1 visualizes the ranking and the relative contributions of the refusal scores. Appendix D lists all the metrics for the top 5 directions.

4.2.1 Effect of Direction Count on JailbreakBench

Figure 2 shows how overall jailbreakability changes as we vary the number of directions k . Even a single direction ($k = 1$) is sufficient to saturate substring-based ASR, while classifier-rated harmfulness (LlamaGuard-2) increases sharply and plateaus around $k = 5$. Increasing k beyond 5 provides negligible benefit. Appendix C contains the category wise ASR evaluations for the Jailbreak Bench complying with the same trend.

Table 2: Attack Success Rate (ASR) on JailbreakBench

Condition	Substring Matching ASR	LlamaGuard2 ASR
Baseline (No Intervention)	22%	16%
$k=1$ Direction Ablation (Direction Removed)	100%	85%
$k=5$ Direction Ablation (Direction Removed)	99%	88%

4.2.2 Baseline, $k=1$, and $k=5$ Steering Comparison

Figure 5 illustrates how jailbreakability when we move from the baseline model to steering with the strongest single direction ($k=1$) and then to the top five directions ($k=5$). Ab-

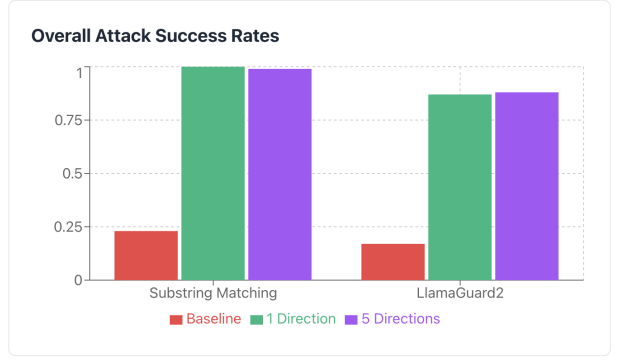


Figure 3: JailBreakBench ASR Analysis

lating only the top-ranked axis (layer 12, position -5) already yields near-maximal substring ASR and substantially increases classifier-rated harmfulness, indicating that a large portion of the jailbreak signal is concentrated in a single latent feature.

Applying the top five directions amplifies these effects. Substring ASR rises from 0.23 (baseline) to 0.99, while LlamaGuard-2 harmfulness increases from 0.17 to 0.88. The corresponding per-category view in Appendix C shows universally saturated ASR across all harmful categories, demonstrating that the discovered directions are robust and generalize across diverse jailbreak types.

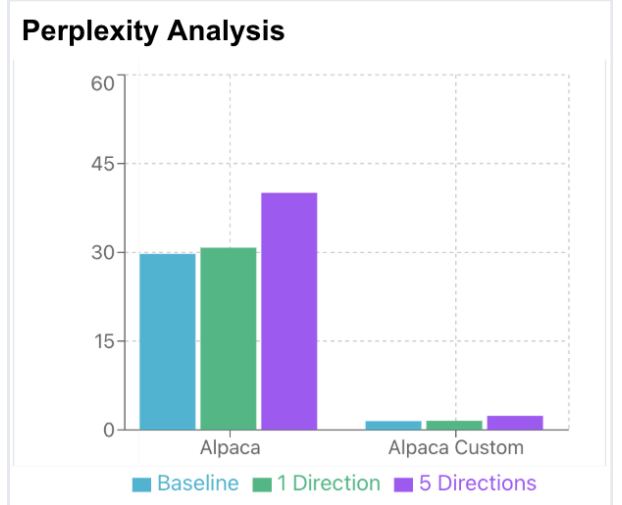


Figure 4: Perplexity Analysis

The Baseline model exhibited strong refusal, with a low ASR of 16%–22%. Directional ablation achieved a near-complete bypass, yielding an ASR of 100% by substring matching and a robust 85% as assessed by the LlamaGuard2 classifier.

4.3 Capability Preservation (Perplexity)

To assess whether refusal-direction ablations degrade core language modeling ability, we evaluate perplexity and cross-entropy on the Alpaca instruction-following dataset (Table 3a). Steering with a single direction introduces only a minimal increase in perplexity relative to the baseline, indicating that the intervention is highly localized and preserves general modeling behavior. However, applying the top five directions leads to a larger rise in both CE loss and perplexity,

suggesting that cumulative ablations begin to interfere with broader linguistic representations. Overall, the results show that small- k steering maintains instruction-following capability, while larger- k interventions trade off language modeling fidelity for stronger control over refusal behavior.

Table 3: Language Modeling Capability Preservation

(a) Alpaca (Instruction Following)		
Condition	CE Loss	Perplexity ($\mathcal{P}$)
Baseline	3.39	29.73
k=1 Ablation	3.43	30.79
k=5 Ablation	3.69	40.06
(b) Alpaca Custom		
Condition	CE Loss	Perplexity ($\mathcal{P}$)
Baseline	0.356	1.43
k=1 Ablation	0.397	1.49
k=5 Ablation	0.424	1.53

4.4 Cost-Efficiency of Refusal Direction Intervention

A key advantage of the Directional Ablation technique is its minimal computational overhead, positioning it as a significantly more accessible and cost-effective attack vector compared to resource-intensive adversarial fine-tuning methods like LoRA or full-parameter updates typically used to break TAR models. The entire refusal direction pipeline, from generation to final selection, demonstrated remarkably low utilization metrics from running on NVIDIA H100 PCIe 80GB:

- **Elapsed Time:** 551.48 s (approx. 9.2 minutes)
- **Average CPU/GPU Utilization:** Low average utilization (CPU avg: 2.5%, GPU avg: 6.2%).
- **Peak GPU Usage:** 94.0% utilization and 20.14 GB max memory, indicating a burst of activity during the forward passes required for vector extraction, but overall low resource commitment.

5. Alternative Intervention Study: Neuron-Level Patching

This section details an alternative investigation into weakening the model’s safety behavior by intervening on individual neurons identified through a change-score methodology, contrasting a Base model (minimal safety training) against the TAR-hardened target model. This study was exploratory and aimed to assess the difficulty of disrupting safety using a neuron-focused, cross-model approach.

5.1 Neuron Ranking and Change Score Pipeline

The methodology for identifying safety-sensitive neurons differed from the direction-based approach by relying on the discrepancy between two related models: the minimal-safety Base model and the high-safety TAR-hardened model. The objective was to identify neurons that shifted most significantly during the hardening process.

5.1.1 Change Score Calculation and Ranking

1. **Forward Passes:** Both the Base model and the TAR model were run on the same set of prompts, and intermediate activations were collected specifically at the post-activation position (MLP output before the residual connection) within each transformer block.
 2. **Change Score Computation:** For every neuron at every layer, a Root Mean Square (RMS) difference was computed between the Base model activations and the TAR model activations over the dataset. This yielded a tensor of shape (layers, hidden dimension).
- $$\text{Change Score}^{(l,n)} = \text{RMS}(\text{Activation}_{\text{TAR}}^{(l,n)} - \text{Activation}_{\text{Base}}^{(l,n)})$$
3. **Ranking:** Neurons were ranked globally from largest to smallest change score. The highest-ranked neurons are those whose behavior shifted most strongly in response to the TAR alignment training.

5.1.2 System Automation

To facilitate iterative experimentation, a robust pipeline was developed, including preprocessing, caching of intermediate tensors for repeated use, automated change score computation, and post-analysis visualization (histograms of high-change neuron locations). This automation allowed for efficient testing of various neuron selections and intervention schemes.

5.2 Neuron Intervention Experiments

Two primary families of neuron-level interventions were implemented using the ranked list of neurons: Zero Ablation (within the TAR model) and Dynamic Patching (transferring activations from the Base model to the TAR model).

5.2.1 Zero Ablation within the TAR Model

Zero ablation served as a direct causal test, mirroring simpler intervention techniques by permanently disabling a selected top fraction (e.g., 0.5%–10%) of the highest-change neurons by setting their activations to zero via a registered permanent hook.

- **Refusal Persistence and Resilience:** A significant finding was the resilience of safety behavior. Even after zeroing a large set of the highest-change neurons, the TAR model frequently continued to refuse harmful questions, indicating that the safety signal is highly distributed and not reliant on only the most-changed neurons.
- **Semantic Degradation:** The intervention did not produce a clean safety toggle. Instead, the model’s language quality often degraded, resulting in repetitive phrases, truncated sentences, or partially corrupted text, even when a refusal was still attempted. This suggests that these highly-ranked neurons encode a mixture of refusal-related features and essential general semantic processing.

5.2.2 Dynamic Patching from Base to TAR

Dynamic patching was a more aggressive, cross-model approach where selected neurons in the TAR model were overwritten with the corresponding instantaneous activation values from the Base model during generation. This aimed to

weaken safety by injecting less aligned behavior into the TAR model’s safety-sensitive neurons.

1. **Donor Run:** The Base model (donor) was executed first to cache its post-activation tensors.
2. **Target Run with Hooks:** The TAR model (target) was subsequently run with a hook at the post-activation sites. The hook function dynamically replaced the target model’s activations for the top-ranked neurons (typically at the last time step) with the cached values from the donor model.

Experimental Results: Patch Strength vs. Coherence
The dynamic patching experiments revealed a consistent pattern defined by the intervention’s strength:

- **Strong Patches (Large Fraction / Wide Temporal Window):** These interventions resulted in catastrophic model failure, including long runs of repeated tokens, strange invented words, and outputs that completely failed to complete the task. This instability was confirmed by control runs (TAR → TAR patching), indicating that the wholesale replacement of internal activations is inherently disruptive to the network’s processing, independent of the alignment difference between models.
- **Weak Patches (Small Fraction / Last Token Only):** When patches were gentle enough to preserve fluency and coherence, the TAR model’s safety stance remained largely unchanged. Refusals persisted in the original wording, or showed only small, stylistic variations, confirming the robustness of the safety circuits against minimal perturbations.
- **Intermediate Settings:** In the narrow middle ground between coherence and catastrophic failure, the model’s refusals became noisier (e.g., mixing refusal with irrelevant content), but rarely collapsed entirely into a fully harmful response.

The conclusion from the dynamic patching study is that neuron-level activation transfer, based solely on the RMS change score between a Base and TAR model, acts as a noisy perturbation rather than a predictable or tunable safety control knob. This contrasts sharply with the surgical efficacy achieved by the Directional Ablation method presented in the main results.

6. Discussion: Asymmetry and Locality in Safety Mechanisms

6.1 Why Inverse Neuron Patching Fails: A Mechanistic Explanation

Prior work has shown that “safety neurons”—neurons whose activations correlate with refusal or safety-oriented behavior—can be transferred from an aligned model to an unaligned base model to partially reproduce aligned behavior. However, the inverse operation (transferring neurons from an unaligned model into an aligned one) behaves very differently. In practice, only a very small number of neurons can be replaced before the model collapses into nonsensical outputs. This section explains why this asymmetry is expected by examining how alignment is implemented inside modern transformer models.

6.2 Representation-Space vs. Parameter-Space Interventions

Two families of interventions are relevant. “Refusal direction” methods operate directly in the model’s internal **representation space** (residual stream activations): they identify a direction in the hidden activations that corresponds to refusal and then push the residual stream away from that direction. Because representations in large models are highly redundant and approximately linear, modifying the state within this space tends to produce smooth and predictable changes to behavior, acting like a global safety switch.

Neuron-patching interventions, in contrast, operate in the **parameter space** (MLP neuron outputs). They physically overwrite activations in the MLP blocks at the post-activation hook point. Transformer blocks, however, are highly coupled systems: even small changes in MLP output perturb the statistical properties of LayerNorm, affect gating, shift activation scales, and propagate non-linearly through every downstream layer. Whereas representation interventions modify the model’s *state* globally and linearly, neuron patching attempts to surgically change the actual *circuits* the model uses to compute, leading to brittle and non-linear effects.

6.3 Alignment Creates Distributed and Irreversible Structure

Post-training alignment methods (e.g., SFT or RLHF) do not simply activate a few safety-related neurons. They reshape entire circuits: attention routing, MLP gating, logit biases, and refusal-specific decoding patterns all adapt jointly. The set of “safety neurons” identified by attribution methods are therefore *correlates* of alignment, not the full causal mechanism.

When neurons are transferred from an aligned model into a base model, they introduce part of a circuit that the base model was already compatible with, so behavior changes in a controlled manner. But when neurons are transferred from a base model into an aligned model, they overwrite components that the aligned model expects to interact with a much larger, safety-hardened subsystem. As a result, their activations become miscalibrated: they amplify or suppress the wrong features, disrupt LayerNorm statistics, and often push activations into regimes the downstream layers were not trained to handle.

6.4 Why Neuron Patching Fails to Invert Safety

Replacing a small number of unaligned neurons in an aligned model typically *increases* refusal because the rest of the safety circuitry dominates. The patched neurons no longer represent the same features that the surrounding layers expect, which causes misfires in downstream safety pathways and results in more conservative behavior.

Replacing a large number of neurons does not remove alignment; instead, it destroys the calibrated structure that alignment training created. Because the unaligned neurons are not functionally compatible with the aligned model’s attention patterns, activation scales, or residual-stream geometry, the model exhibits unstable behavior—often producing

incoherent or degenerate outputs, as observed in our "Strong Patch" experiments.

6.5 Comparative Analysis and Conclusion

The key insight is that alignment modifies the model in a *distributed, architecture-wide* way. Safety-related behavior is not localized to a few neurons; it is implemented through coordinated changes across many layers. The Refusal Direction attack succeeds because modern alignment methods (such as TAR) primarily introduce safety behavior via low-rank modifications of late-layer representations. These modifications manifest as a relatively clean linear direction that is easily counteracted.

- **Refusal Direction (Success):** This intervention works by modifying the linear, redundant *representation space* and is applied broadly across **all components** (Attention and MLP) in **all layers**. As demonstrated by our experiments (Section 2.5), it is **bidirectional**, working both to disable safety (Ablation) and induce it (Activation Addition).
- **Neuron Patching (Failure):** This intervention attempts to change the non-linear *parameter space* and is limited to only the **MLP components** (the post-activation hook point). Because the safety signal is highly distributed, attempting to disable it by swapping out only the most-changed neurons causes computational collapse rather than controlled bypass. Our experiments confirm it is **unidirectional** (only Base $\rightarrow$ TAR transfer, making it safer, is tolerated by the model, not the inverse).

In short, the refusal-direction intervention works cleanly because it perturbs the model state in a smooth, low-rank, linear subspace that the model’s circuits can easily interpolate away from. Neuron-based interventions modify the underlying circuitry in a brittle, high-rank, and highly nonlinear manner, leading to model instability. This explains the profound asymmetry in efficacy between the two methods.

7. Related Work

7.1 Mechanistic Interpretability and Activation Steering

Our work builds upon the growing field of mechanistic interpretability, specifically the hypothesis that high-level features are encoded as linear directions in the activation space of LLMs [1, 4]. Ardit et al. [1] demonstrated that refusal in standard chat models is mediated by a single direction, and that ablating this direction can bypass safety guardrails. Our work extends this finding to the domain of "durable" safeguards, showing that even models explicitly trained to resist weight tampering (TAR) still rely on this linear refusal mechanism. This contrasts with earlier activation steering methods that primarily focused on steering sentiment. [4].

7.2 Durable Safeguards & Tampering Resistance

The primary target of our investigation is the Tampering Attack Resistance (TAR) framework proposed by Tamirisa et al. [2]. TAR aims to create open-weight models that retain safety properties even after adversarial fine-tuning. It

achieves this via a meta-learning objective that minimizes a safety loss (e.g., negative entropy on hazardous prompts) after a simulated fine-tuning step. While TAR represents a significant advance over standard unlearning methods like RMU (Representation Misalignment) or RepNoise, which are easily reversed [2], our results show that TAR’s robustness is largely confined to gradient-based optimization attacks. By modifying the model at inference time via directional ablation, we bypass the optimization landscape entirely.

7.3 Vulnerabilities of Safeguards

Recent work by Qi et al. [3] has critically examined the durability of safeguards like TAR and RepNoise. They demonstrated that these defenses can be brittle to variations in attack configuration, such as the use of LoRA versus full fine-tuning, or changes in learning rate schedules (e.g., cosine decay). Furthermore, they showed that "unlearned" knowledge could often be recovered by simply changing the prompt template or fine-tuning on benign datasets, suggesting that the hazardous knowledge was masked rather than erased. Our work complements these findings by introducing a distinct class of attack: while Qi et al. focused on optimizing the weights (fine-tuning), we show that a computationally cheap, training-free intervention on the activations is sufficient to break the safeguard. This reinforces the conclusion that current "unlearning" methods do not structurally excise hazardous capabilities from the model.

8. Conclusion

In this work, we demonstrated that the "durable" safety guarantees of the TAR-Bio v2 model rely on a linear refusal direction that persists despite robust training. By identifying and ablating this direction, we achieved an Attack Success Rate of greater than 85% on harmful prompts without performing any fine-tuning. This result challenges the prevailing definition of "tamper resistance," which has largely focused on weight stability against gradient descent. We conclude that true safety durability requires not just resistance to optimization, but a fundamental restructuring of the model’s internal representations to prevent simple linear bypasses. Future work in robust alignment must move beyond weight-hardening to address the geometric persistence of refusal features in activation space.

References

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A. Contributions

- Sania implemented refusal direction extraction, ablation/ActAdd experiments, and large-scale evaluation across the JailbreakBench. Led analysis on ASR, perplexity, and low-compute efficiency findings.
- Stephanie handled dataset preprocessing and harmful/harmless split logic, plus direction ranking, false-positive evaluation, and steering/heatmap visualizations.
- Adyasha designed neuron-level intervention strategy using change-score ranking. She also analyzed and documented the mechanistic interpretability underlying both approaches.
- Zhirui built tooling for neuron selection and activation logging, and conducted threshold, layer-sweep, and patching evaluations to identify sparse vs. distributed safety representations.

B. Algorithm for Tamper Resistance(TAR)

Algorithm 1 TAR: Tampering Attack Resistance

Require: Initial LLM parameters θ_0 , train-time adversary set $\mathcal{A}_{\text{train}}$, capabilities_metric proxy dataset $\mathcal{D}_{\text{retain}}$, safety_metric proxy dataset $\mathcal{D}_{\text{TR}}$, outer steps N , learning rate η , number of sampled adversaries K , tamper-resistance loss scale λ_{TR} , retain loss scale λ_{retain} , $h_{\theta}(\cdot)$ returns the residual stream hidden states for model parameters θ .

- 1: Apply Initial Safeguard to $\theta_0 \rightarrow \theta_{G_0}$
- 2: **for** $i = 1$ to N **do**
- 3: $g_{\text{TR}} \leftarrow 0$ $\triangleright$ For accumulating tamper-resistance gradient
- 4: Sample $x_{\text{TR}} \sim \mathcal{D}_{\text{TR}}$
- 5: **for** $k = 1$ to K **do**
- 6: Sample attack $\sim \mathcal{A}_{\text{train}}$ $\triangleright$ Tamper-resistance loss from Equation 1
- 7: $g_{\text{TR}} \leftarrow g_{\text{TR}} + \frac{1}{K} \nabla_{\theta_{i-1}} \mathcal{L}_{\text{TR}}(\text{attack}(\theta_{i-1}), x_{\text{TR}})$
- 8: **end for**
- 9: Sample $x_r \sim \mathcal{D}_{\text{retain}}$ $\triangleright$ RepE retain loss from Equation 2
- 10: $g_{\text{retain}} \leftarrow \nabla_{\theta_{i-1}} (\mathcal{L}_{\text{LM}}(\theta_{i-1}, x_r) + \|h_{\theta_{i-1}}(x_r) - h_{\theta_{G_0}}(x_r)\|_2^2)$ $\triangleright$ Full tamper-resistance update
- 11: $\theta_i \leftarrow \theta_{i-1} - \eta(\lambda_{\text{TR}} \cdot g_{\text{TR}} + \lambda_{\text{retain}} \cdot g_{\text{retain}})$
- 12: **end for**
- 13: $\theta_G \leftarrow \theta_N$
- 14: **return** θ_G

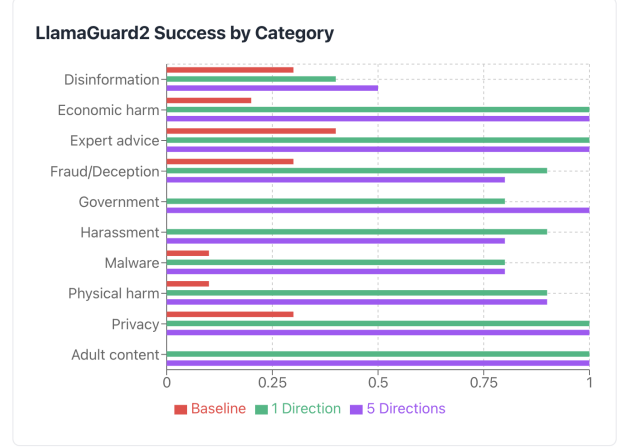


Figure 5: JailBreakBench ASR Analysis by Category for baseline, k=1 and k=5

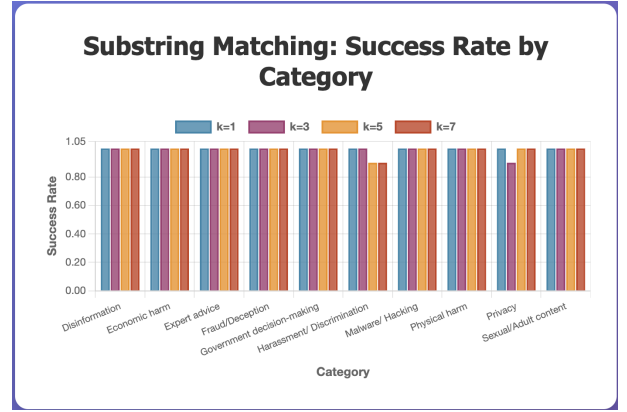


Figure 6: Substring Matching: Success Rate by Category on JailBreak Bench across various k

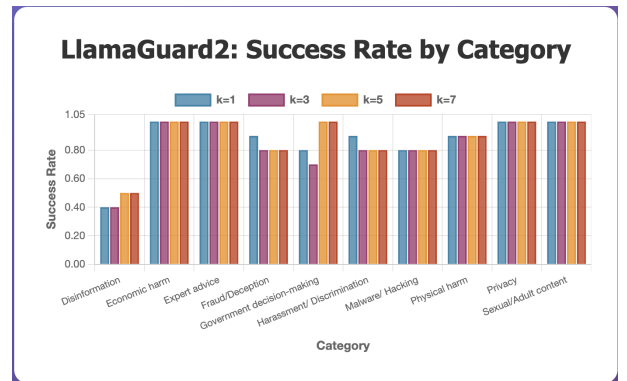


Figure 7: LlamaGuard2: Success rate by Category on Jail-Break Bench across various k

C. Category-wise JailBreak Bench ASR

D. Top-k Refusal Direction Metrics

Top Steering Directions (Ranked)					
Rank	Layer	Position	Refusal Score	Steering Score	KL Divergence
1	12	-5	-7.5430	0.5062	0.0145
2	12	-1	-7.1104	0.4616	0.0177
3	13	-5	-6.9424	0.3261	0.0288
4	11	-5	-6.0300	0.2668	0.0076

Figure 8: Top Steering Directions Ranked